

Model Performance and Feature Analysis: A Pipeline Approach

1 Introduction

This project focuses on forecasting soil moisture content (SWC) using a variety of machine learning models. I developed a modular forecasting pipeline to evaluate model performance across different temporal configurations and soil depths. The models tested include convolutional neural networks (CNNs), long short-term memory networks (LSTMs), ARIMA, and XGBoost.

The pipeline supports varying input/output window sizes, feature engineering and filtering, seasonal testing splits, hyperparameter tuning, and manual and automatic feature selection.

I consistently evaluated performance using Mean Absolute Error (MAE). Experiments were conducted for 10 cm SWC, using hourly meteorological and soil measurements from multiple stations in Texas. Models were tested for performance with 48 hour, 72 hour, and 168 hour prediction targets. I also tested performance by season, and also attempted testing the affect of manual hyperparameter tuning on these large models, as is standard practice with simpler ML models through techniques like GridSearchCV. At the end, I presented some results from my feature importance investigation with these various models.

2 Dataset

I used hourly meteorological and soil data from six stations in Texas. The dataset includes:

- Target variables: SWC at 5, 10, 20, and 50 cm depths (SWC_5, SWC_10, etc.).
- Environmental features: Soil temperature at multiple depths (T_5, T_10, ..., T_50), precipitation (Ppt), relative humidity (RH), air temperature (Tair), solar radiation (Srad), wind speed, wind direction.
- Time encodings.
- Location Data (Latitude/Longitude)

Missing values and anomalies have been previously addressed and were not a concern, as this project was built on a cleaned dataset. Location data was excluded to avoid unnecessary/extraneous complexity.

3 Feature Filtering and Engineering

3.1 Feature Filtering

To streamline model input and reduce noise, we applied both manual feature selection and optional correlation-based filtering.

Manual Selection:

- Many features are collected at multiple depths (e.g., SWC_5, SWC_10, SWC_20, T_5, T_50).
- We retained:
 - SWC_10 - as the primary prediction target.
 - T_5 - due to strong correlation and consistent model performance.

Automated Correlation Filtering (Optional):

- A correlation matrix is computed for all numeric features not manually retained.
- Features with pairwise correlation above a set threshold are flagged.
- In each pair, one feature is dropped to prevent multicollinearity and redundancy.
- This reduces input complexity and helps mitigate overfitting.

Final Selection:

- Core features: SWC_10, T_5
- Optional additions based on analysis:
 - RH, Srad, Ppt, Wind_X, Wind_Y
- These were selected based on exploratory analysis (`data_analysis.ipynb`).
- Most tests were run using only SWC_10 and T_5.
- In the *Feature Importance* section, we explore the impact of adding more features.

3.2 Feature Engineering

- **Wind Vectors:** Wind_X and Wind_Y are Cartesian transformations of Windspeed and Winddirection, capturing wind velocity as a vector.
- **Temporal Cyclical Features:** DaySin, DayCos, YearSin, YearCos encode seasonality in a continuous, periodic manner using sine and cosine transformations. These features preserve cyclicity and scale values between 0 and 1.

4 Data Windowing and Normalization

The following preprocessing steps are applied to models that require time-series batching and scale-sensitive input representations, specifically CNN, LSTM, Bi-LSTM, and XGBoost. Neural networks in particular are sensitive to input distributions and benefit from windowed inputs.

4.1 Data Windowing

Time series data was transformed into overlapping input-output sequences to enable supervised learning.

- Each input window represents a fixed-length segment of historical data (72 hours of hourly data).
- The model is trained to predict a future target at a fixed offset from the end of the window.
- We evaluated three prediction horizons:
 - 48-hour forecast
 - 72-hour forecast
 - 168-hour forecast
- This process enables multi-step forecasting and generates a large training set from continuous time series data.

4.2 Data Normalization

- All features were scaled using `StandardScaler` to have zero mean and unit variance.
- The transformation was fit on the training data and applied to both train and test splits.
- During model evaluation, only the target variable (`SWC_10`) was inverse-transformed to recover predictions in physical units (soil moisture percentage).
- This ensures comparability across models and accurate interpretation of results.

5 Modeling

5.1 ARIMA (Autoregressive Integrated Moving Average)

ARIMA is a traditional statistical model suited for univariate time series. It was used as a baseline in our experiments. Since ARIMA cannot utilize exogenous inputs by default, it generally underperformed compared to more complex models. It was most effective in winter, when soil moisture patterns were stable since rain was little, but failed to capture multivariate and non-linear dependencies in other seasons. It achieved its best performance in more stable conditions but struggled to capture multivariate dependencies, leading to the highest RMSE overall.

5.2 ARIMAX (ARIMA with Exogenous Inputs)

ARIMAX extends ARIMA by incorporating exogenous variables, enabling it to use additional predictors such as temperature, rainfall, and radiation. This provided it with more contextual information than ARIMA alone, improving performance slightly better than ARIMA, but still trailing behind modern ML models. Its linear nature limited its ability to handle complex feature interactions, leading to a relatively high RMSE.

5.3 LSTM (Long Short-Term Memory Networks)

LSTM is a deep learning model designed for sequence data. It was implemented using Keras, with parameters like number of units, activation function, and dropout rate set manually or through tuning. LSTM performed well in seasons with consistent temporal patterns (especially winter), but its performance degraded a bit in noisier seasons like summer. It showed a strong ability to model long-term dependencies, but was sensitive to hyperparameters and required normalization and careful windowing.

5.4 BiLSTM (Bidirectional Long Short-Term Memory)

BiLSTM processes time series in both forward and backward directions, enabling it to learn from both past and future context within a training window. This gave it an edge over standard LSTM in capturing symmetrical dependencies, making it the second-best performer in terms of RMSE. BiLSTM was especially effective when short-term fluctuations influenced outcomes as well as long-term ones. Its strong performance suggests that bidirectional context is valuable for soil moisture forecasting, as well as forecasting complex time series data in general.

5.5 CNN (Convolutional Neural Network)

CNNs were used with 1D convolutions to extract temporal features over short windows. While CNNs performed well in stable conditions (winter), their performance dropped in seasons with more variability. CNNs require careful design of filter sizes and dropout to prevent overfitting. Unlike LSTM or BiLSTM, CNN models are more local in focus and may miss long-range dependencies.

5.6 XGBoost (Extreme Gradient Boosting)

XGBoost consistently delivered strong performance across all seasons. It is robust to outliers, can handle non-linear patterns, and performs well without needing heavy data preprocessing. By flattening the input window into a 2D array, XGBoost was able to capture relationships across time steps effectively. It outperformed all other models in terms of both accuracy and consistency, with low RMSE, MAE, and MAPE across the board.

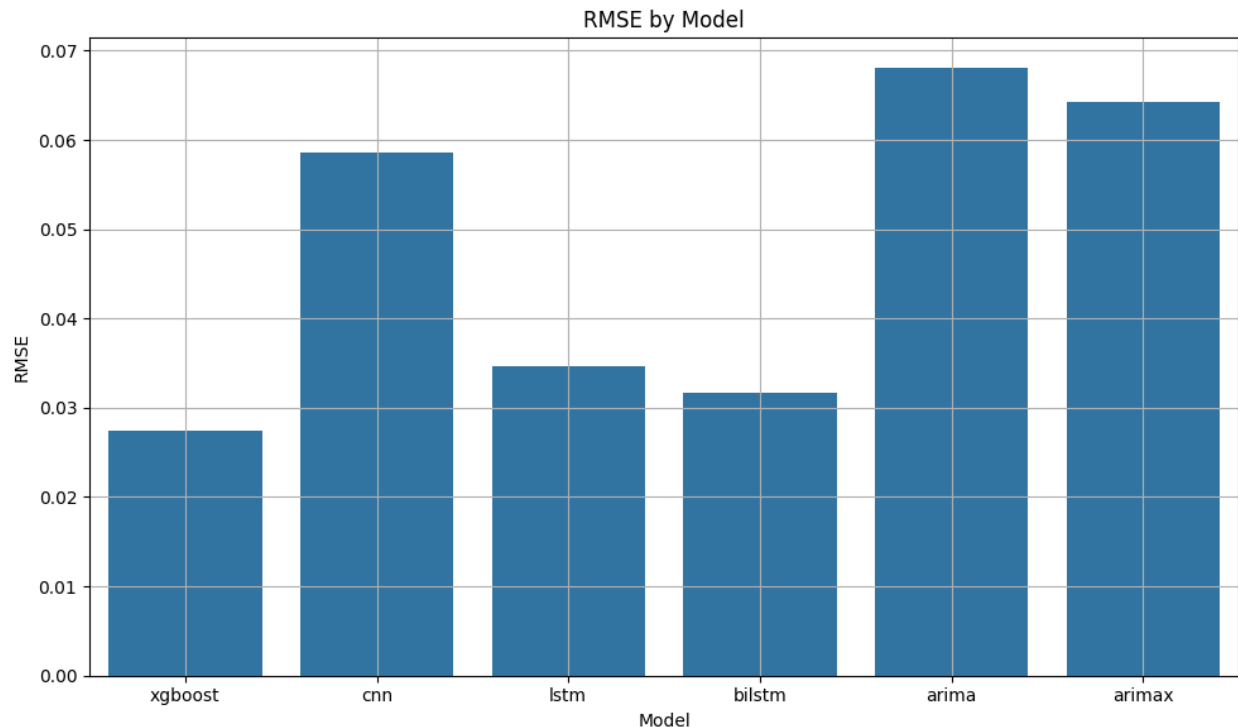


Figure 1: Overall RMSE comparison across models aggregated over all seasons and horizons.

6 Seasonal Performance

Model performance varies noticeably across seasons. All models perform best in winter, likely due to little rain and more steady weather patterns. In contrast, summer and fall are more challenging due to complex interactions of precipitation and heat. Among the models, XGBoost demonstrates strong and stable performance across all seasons, while BiLSTM also delivers consistently strong results, especially in more variable conditions. LSTM performs well but is more sensitive to seasonal fluctuations that are difficult to capture. CNNs show good winter performance but suffer more in irregular conditions. ARIMA consistently lags behind due to its linear, univariate nature, while ARIMAX benefits from exogenous features and performs a bit better than ARIMA.

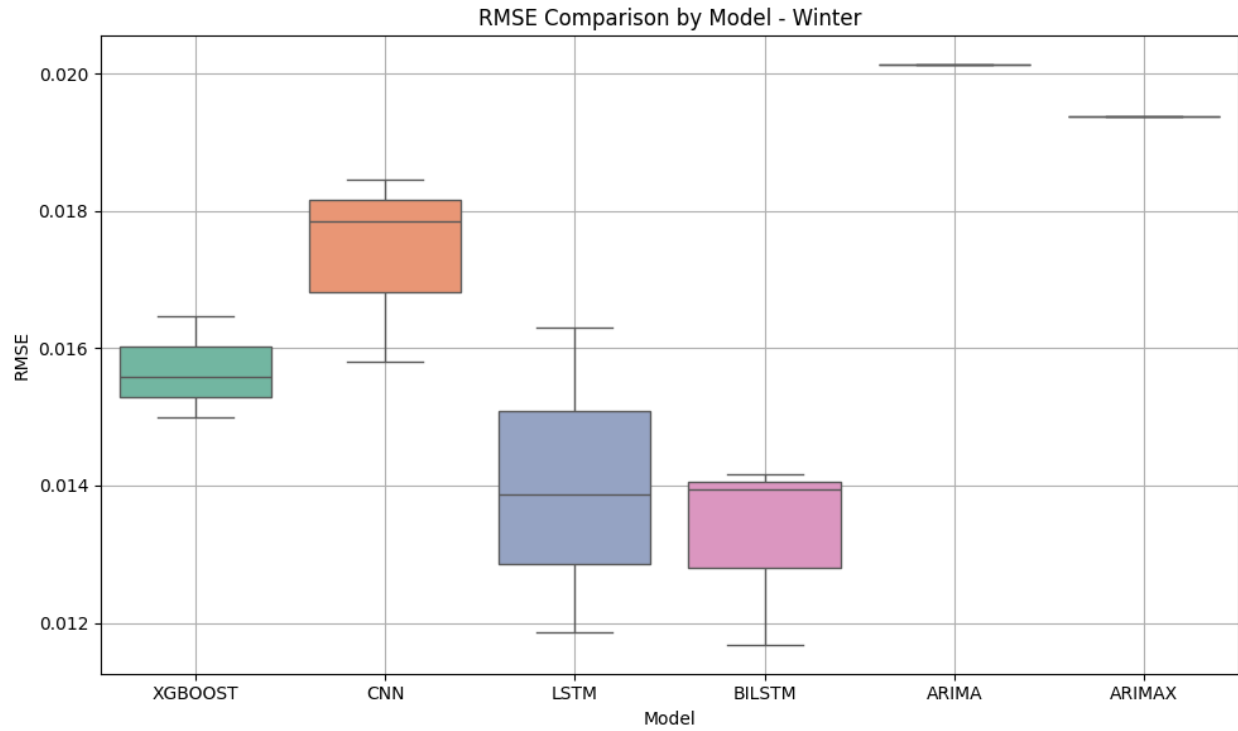


Figure 2: Seasonal model performance (RMSE) - Winter.

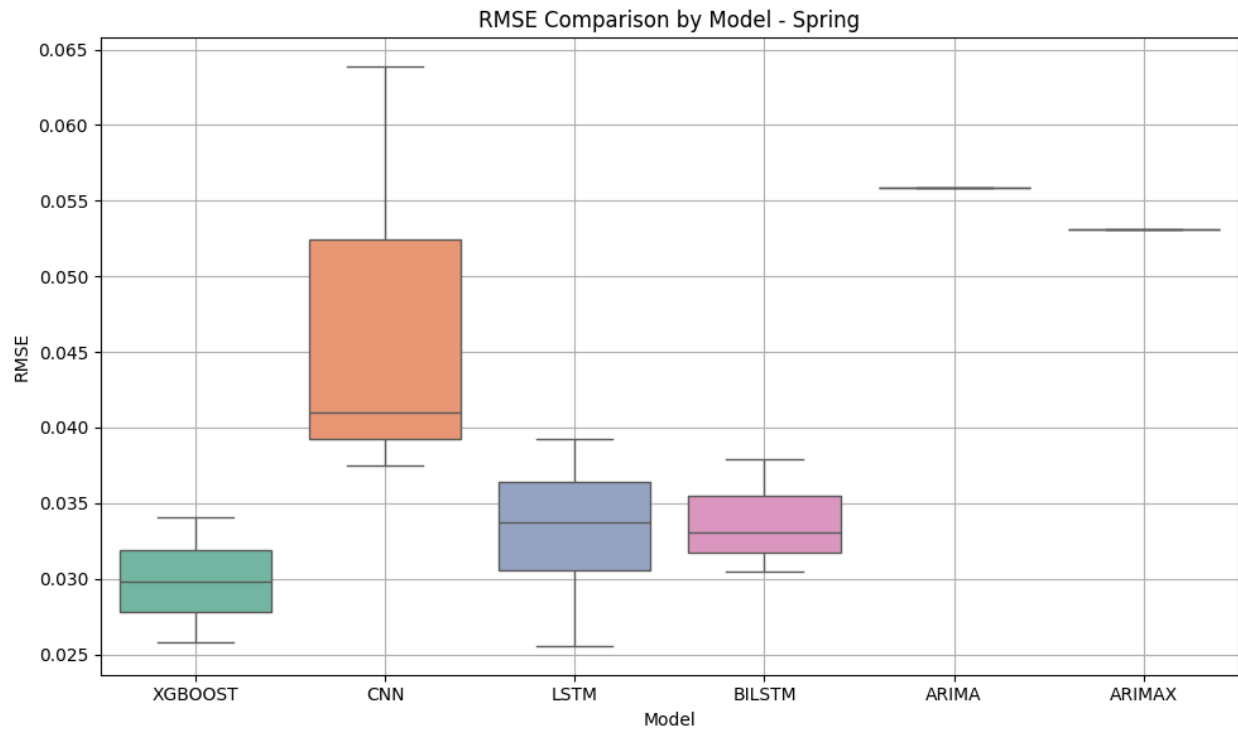


Figure 3: Seasonal model performance (RMSE) - Spring.

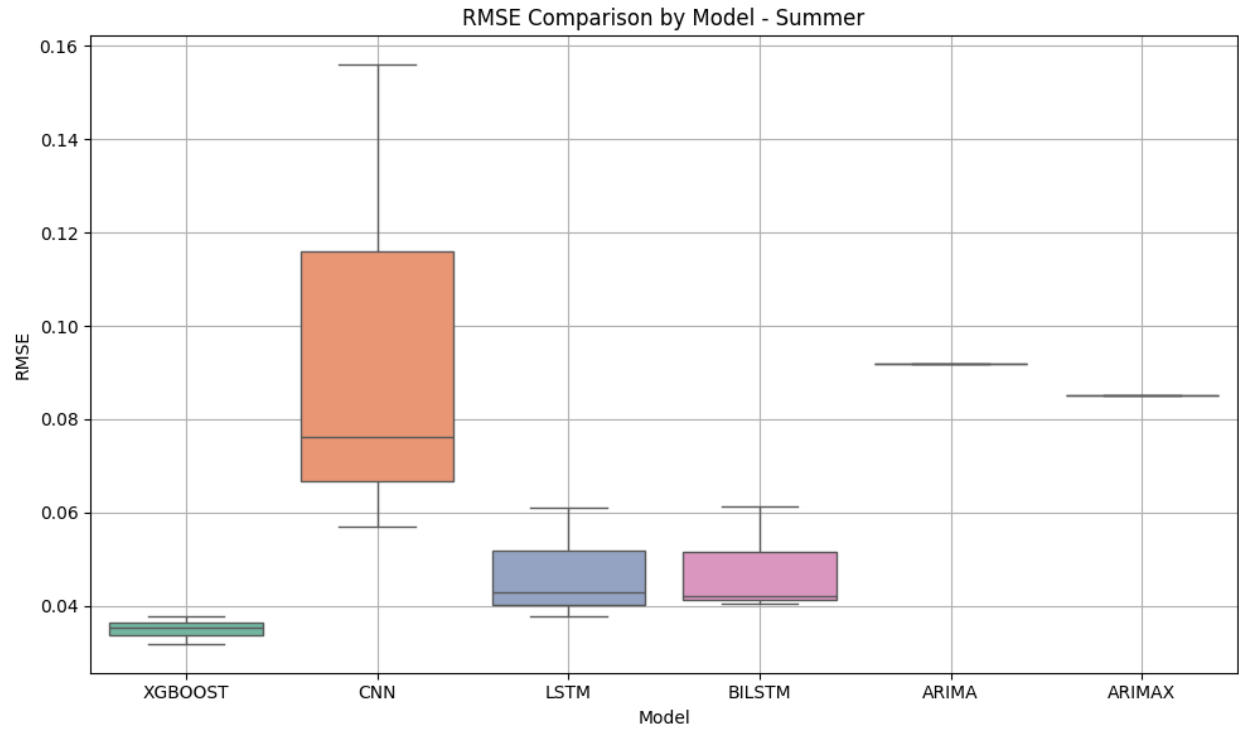


Figure 4: Seasonal model performance (RMSE) - Summer.

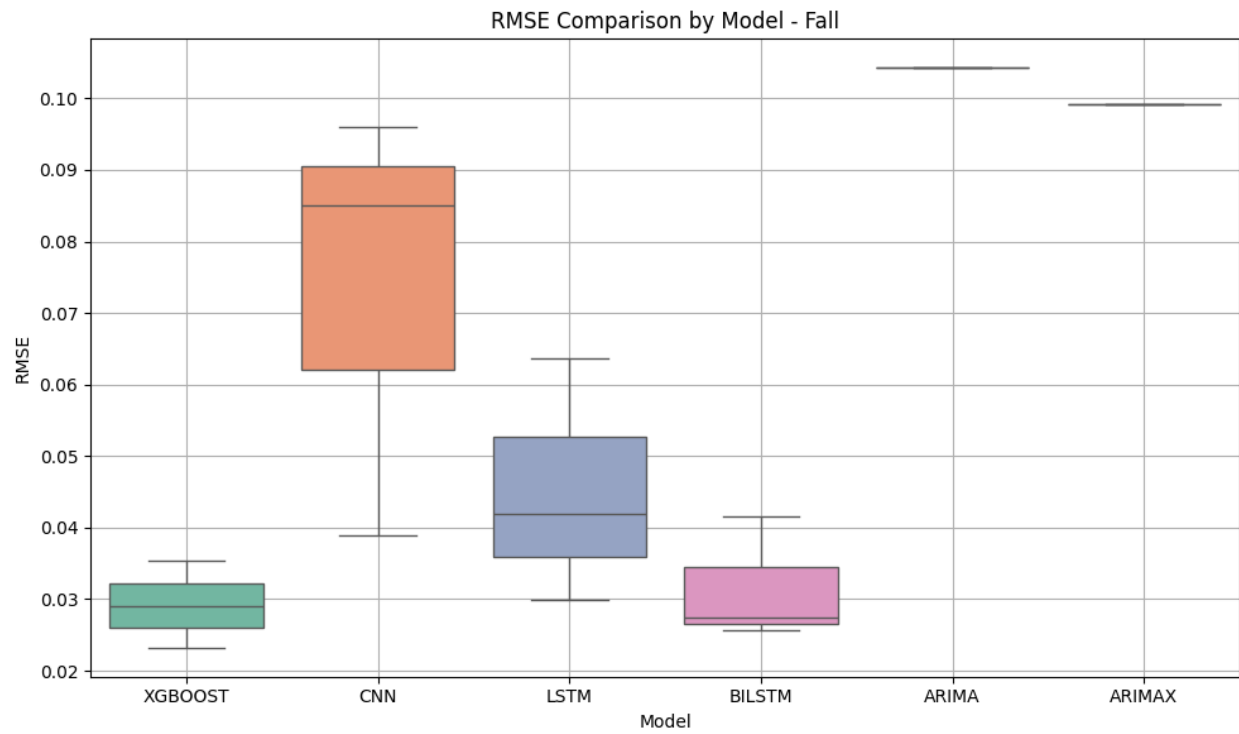


Figure 5: Seasonal model performance (RMSE) - Fall.

6.1 Winter

- All models performed best in winter, which had the lowest average RMSE and MAPE.
- Patterns were more consistent due to lower precipitation variability.
- ARIMA and ARIMAX, despite being linear, performed relatively well, with ARIMAX outperforming ARIMA due to additional predictors.
- XGBoost and BiLSTM achieved the best scores across all metrics, with LSTM close behind.

6.2 Spring

- Performance began to diverge across models due to increased variance.
- CNN and LSTM remained effective but started showing more error spread.
- BiLSTM maintained strong performance, showing better handling of variance.
- ARIMAX generally outperformed ARIMA.
- XGBoost remained stable and continued outperforming others.

6.3 Summer

- This was the most challenging season, with highly variable weather and soil moisture due to rainfall and heat.
- All models showed increased RMSE and MAPE.
- LSTM and CNN had significantly lower correlation, indicating poor trend capturing.
- BiLSTM performed better than LSTM, benefiting from bidirectional context.
- ARIMAX also struggled with capturing complex patterns.
- XGBoost maintained reasonable error rates and high correlation, outperforming others.
- ARIMA performed the worst, showing its limitations with non-stationary data.

6.4 Fall

- Similar to summer, with high variance and lower correlation in LSTM and CNN.
- BiLSTM again proved more robust than LSTM, ranking among the top two models.
- ARIMA showed its worst performance in fall across all metrics.
- XGBoost again remained consistent, highlighting its robustness.

7 Varying Offset Performance

All models showed increasing RMSE with longer horizons, reflecting the logical growing uncertainty with longer-term predictions.

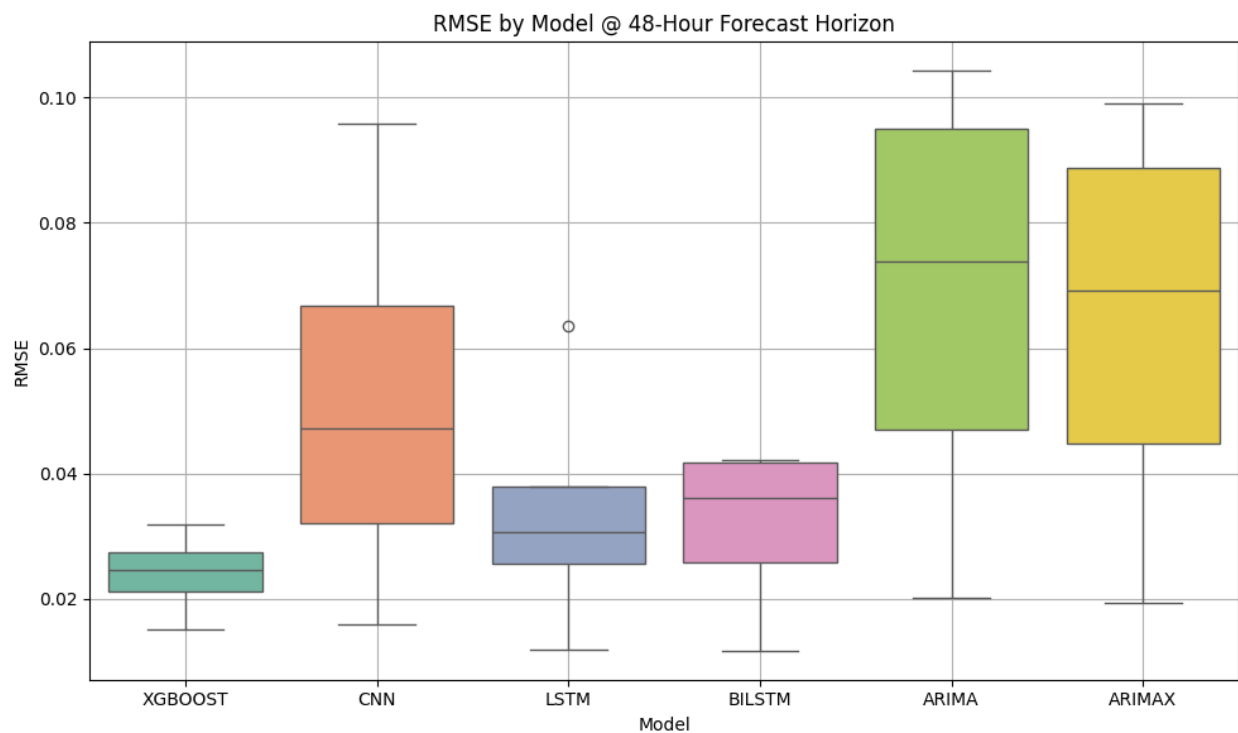


Figure 6: RMSE by model at 48-hour forecast horizon.

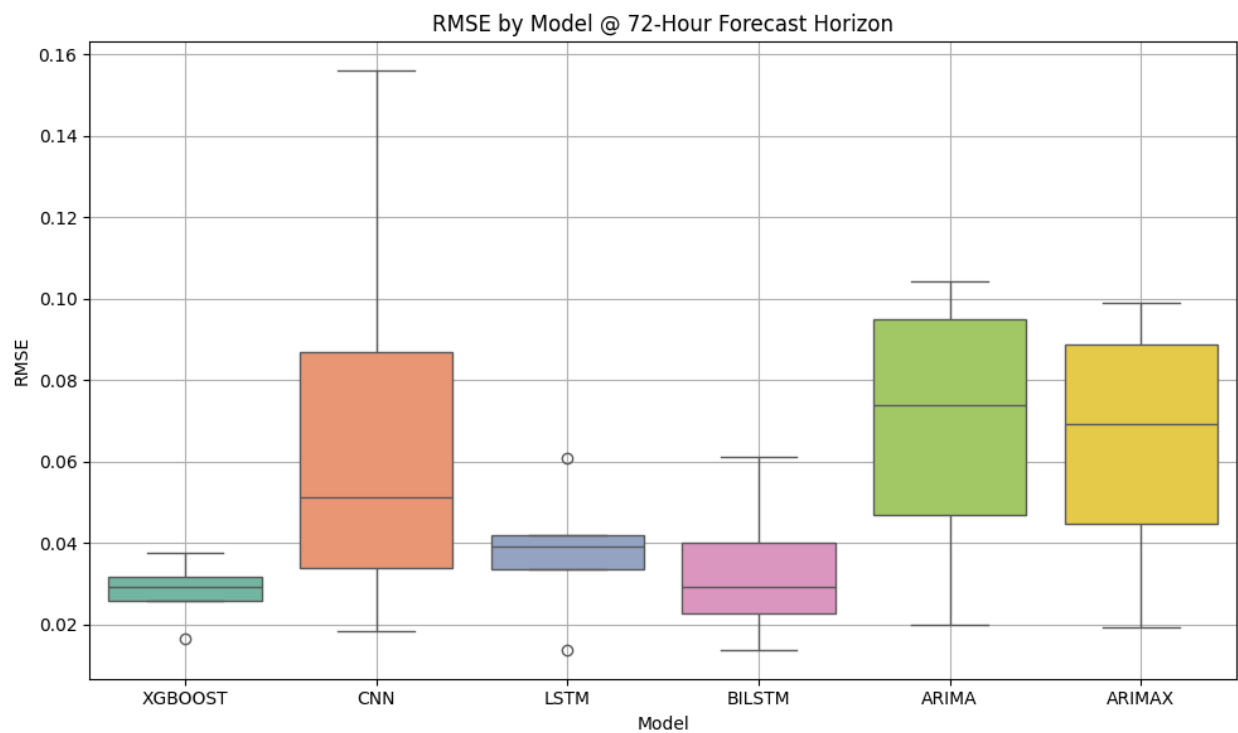


Figure 7: RMSE by model at 72-hour forecast horizon.

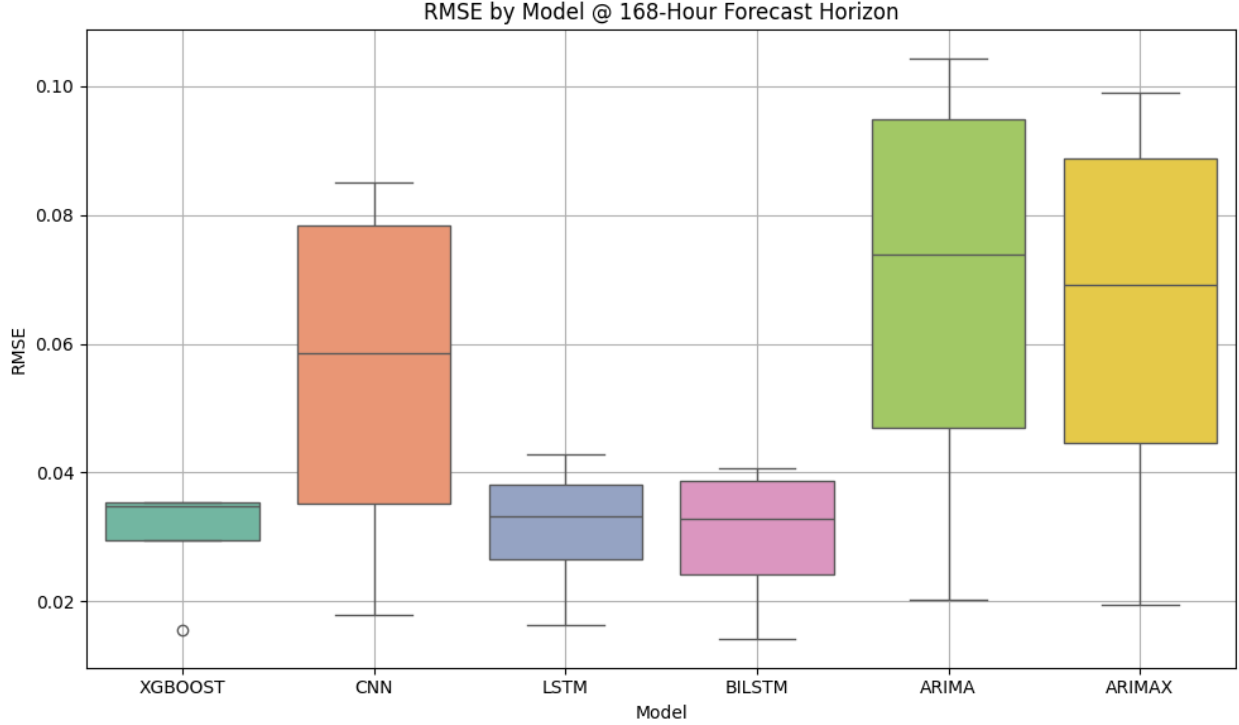


Figure 8: RMSE by model at 168-hour forecast horizon.

- **XGBoost** maintained the most stable RMSE, with relatively small increases even at 168 hours. This demonstrates its robustness in multi-step forecasting.
- **LSTM** and **BiLSTM** consistently ranked among the top performers at every horizon. Their RMSE barely rose with forecast length, showing the effectiveness of memory-based capturing of temporal patterns.
- **ARIMA** and **ARIMAX** performed poorly across all horizons, with the highest RMSEs and the most degradation from 48 hours to 168 hours. Their performance decline confirms their limitations due to a linear nature and difficulty handling compounding uncertainty.

8 Results

8.1 Overall Model Performance

Across all seasons and horizons (Figure 1), XGBoost achieved the lowest aggregate RMSE, closely followed by BiLSTM. LSTM and CNN formed the middle tier, while ARIMAX and ARIMA consistently trailed behind. The ranking was consistent across both error magnitude and stability, proving how model choice directly impacts predictive accuracy in soil moisture forecasting.

8.2 Seasonal Analysis

Model performance varied noticeably from season to season (Figures 2–5). In **winter** (Figure 2), all models delivered their strongest results. The stable, low-variability conditions favored even the simpler statistical approaches, with ARIMAX edging out ARIMA thanks to its additional predictors. XGBoost and BiLSTM led the pack, with LSTM performing closely behind. In **spring** (Figure 3), increasing variability began to separate the models. XGBoost continued to lead, while BiLSTM maintained strong, consistent performance. CNN and LSTM remained competitive but showed more spread in their errors, and ARIMAX maintained

a small advantage over ARIMA. In **summer** (Figure 4), high variability from heat and rainfall created a tougher prediction environment. All models saw higher errors, but the drop in correlation for LSTM and CNN was especially sharp. BiLSTM proved more resilient, while ARIMAX struggled alongside ARIMA. XGBoost handled the conditions best, with only a modest increase in error. In **fall** (Figure 5), the challenges looked similar to summer. XGBoost again remained steady, BiLSTM continued to outperform LSTM, and ARIMA recorded its weakest seasonal performance.

8.3 Horizon Analysis

As expected, forecast accuracy declined as the prediction horizon increased (Figures 6–8). At **48 hours** (Figure 6), XGBoost, BiLSTM, and LSTM, produced similarly competitive results, showing strong short-term forecasting ability. At **72 hours** (Figure 7), the performance gap widened, with XGBoost pulling ahead and BiLSTM holding a clear second place. At **168 hours** (Figure 8), the differences became more pronounced. XGBoost’s error rose only modestly, highlighting its robustness for long-range forecasts. LSTM’s accuracy dropped more sharply than BiLSTM’s, underscoring the benefit of bidirectional context. ARIMA and ARIMAX saw the steepest declines, reaffirming the limitations of these linear models when dealing with compounding uncertainty over extended horizons.

9 Conclusion

This study highlights how strongly model choice, feature selection, and forecast horizon affect soil moisture prediction accuracy. Among all approaches tested, XGBoost stood out as the most reliable and accurate, maintaining strong performance across seasons and forecast lengths, even under highly variable conditions. BiLSTM followed closely, showing particular strength in complex seasonal patterns and demonstrating resilience in both medium- and long-range forecasts. LSTM also performed well for shorter horizons but proved more sensitive to changes in seasonality and forecast length.

ARIMAX, while a clear improvement over baseline ARIMA thanks to its use of exogenous inputs, remained constrained by its linear design. Both ARIMAX and ARIMA struggled to maintain accuracy in longer-range and more variable conditions. CNNs proved effective at capturing stable, local patterns but lacked the ability to model long-term dependencies, making them less competitive in volatile seasonal environments.

Overall, the results suggest that tree-based ensemble methods like XGBoost and advanced sequence models like BiLSTM are strong candidates for operational soil moisture forecasting. Future work could explore hybrid models that combine the feature extraction capabilities of deep learning with the structured decision-making of ensemble methods, as well as adaptive systems that adjust the model choice dynamically based on seasonal or meteorological conditions.